

F I N A L R E S U L T S

KPI Robustness in On-Time Performance Measurement

How sensitive is OTP to defensible methodological choices?

Tobias Bussiek

West Chester University of Pennsylvania

April 2026



OTP is reported as a single percentage.

But computing OTP from public GTFS-RT data requires methodological choices and reasonable choices produce dramatically different numbers.

1

Hidden choices

Computing OTP from GTFS-RT requires picking lateness windows, handling early arrivals, selecting stops, and deduplicating overlapping real-time predictions.

2

Invisible upstream

Some choices are recognizable as methodology (e.g., the 5-minute window). Others happen earlier, during stop-event reconstruction, and rarely appear in published reports.

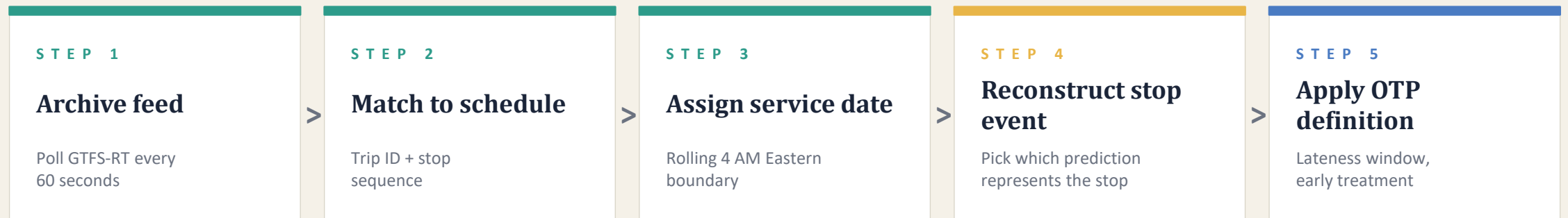
3

Same data, different number

Across reasonable choices, the same underlying data can be reported as 20% on-time, 99% on-time, or anywhere in between. The reader cannot tell which methodology produced the number.

GTFS-RT pipeline choices are part of KPI methodology

Each step in reconstructing a stop event from real-time predictions involves a methodological choice.



STEP 4 GETS MUCH LESS ATTENTION

Snapshot selection: which prediction represents the stop event?

GTFS-RT is a stream of overlapping live predictions, not a finished historical record.

A single scheduled stop appears in many snapshots as the bus approaches. The analyst must pick one.

This study compares three rules: **closest** (prediction nearest scheduled time), **last_before** (last prediction before scheduled time), and **latest** (chronologically last, treated here as the closest GTFS-RT-only proxy).

Six specification dimensions, fully crossed

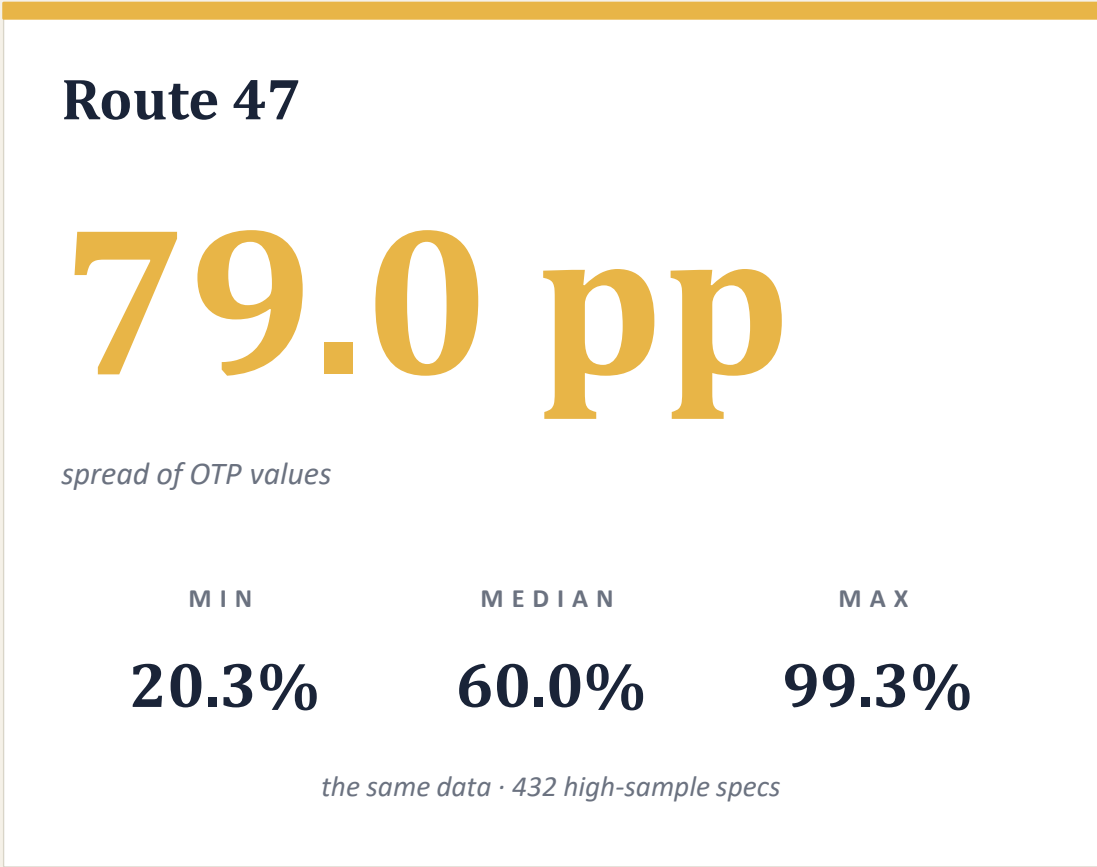
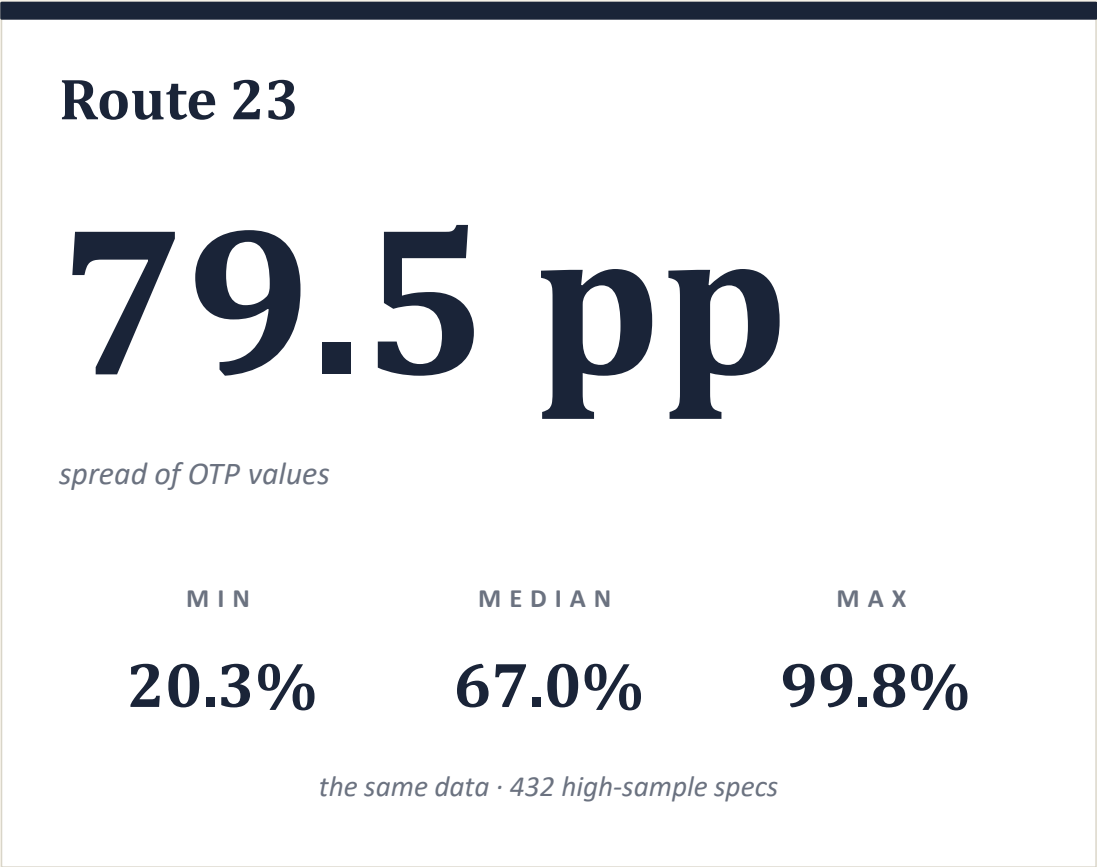
Adapting specification-curve and multiverse analysis (Simonsohn et al. 2020; Steegen et al. 2016) to descriptive KPI measurement.

Dimension	Variants	What it controls
Snapshot rule	closest · last_before · latest	Which prediction represents each stop event
Lateness window	0/+3 · -1/+5 · -2/+7	Schedule-deviation range counted as on time
Early treatment	count_early_as_late · ignore_early	Whether early arrivals count as failures
Stop selection	all_stops · timepoints · terminals	Which stops contribute to route-level OTP
Time of day	AM peak · midday · PM peak · evening · all	Time window of analysis
Delay basis	arrival · departure	Scheduled time used as the reference

$3 \times 3 \times 2 \times 3 \times 5 \times 2 = 540$ specifications per route · **1,080 total across both routes**

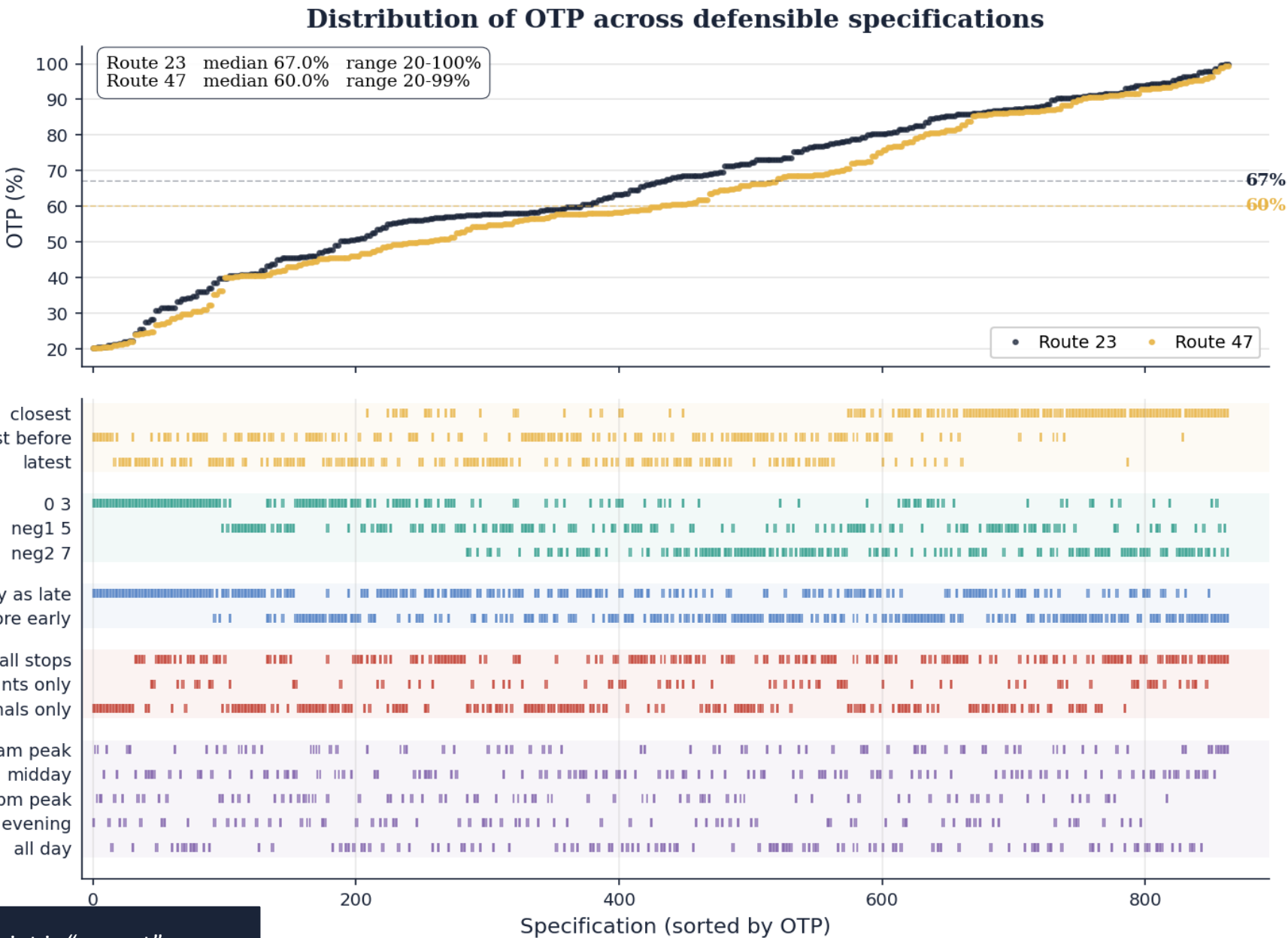
Reasonable choices produce ~80 percentage points of variation

Distribution of OTP across high-sample specifications, 432 per route.



The specification curve

Each point is one high-sample OTP estimate, sorted from lowest to highest.

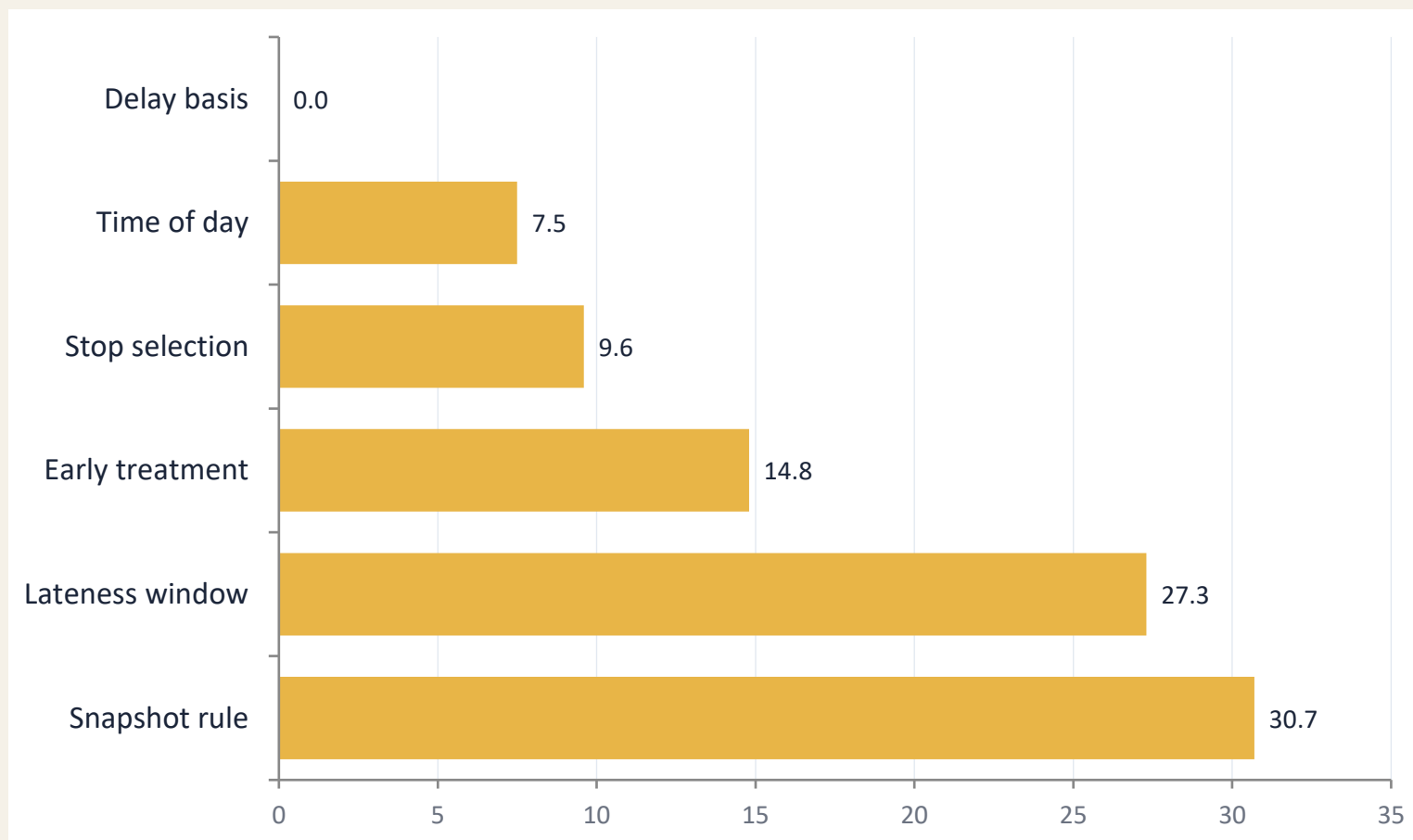


TAKEAWAY The curve is the answer. No single point is “correct”.

Note: delay_basis dimension omitted — both variants produce identical OTP in SEPTA data

Sensitivity by specification dimension

Mean OTP range when varying each dimension while holding all others fixed.



THE FINDING

Snapshot selection is largest

30.7 pp mean range

Largest dimension, even above the well-studied lateness window.

Less visible because it happens upstream during event reconstruction.

Snapshot selection alone shifts OTP by 30+ percentage points

All other dimensions held fixed at one reference specification (−1/+5 window, count early as late, all stops, all day, arrival).

Snapshot rule	What it does	Route 23	Route 47
closest	Prediction nearest scheduled time	87.4%	86.2%
last_before	Last prediction before scheduled time	60.7%	55.9%
latest	Chronologically last prediction (GTFS-RT-only proxy)	56.0%	48.8%

WHY THIS HAPPENS — SELECTION BIAS

The **closest** rule selects the prediction nearest scheduled time. OTP rewards closeness to scheduled time.
The selection rule and the metric measure overlapping properties since the rule biases toward predictions that already look on time.

Extreme values come from combinations of methodological choices

Neither endpoint is the single 'correct' value since each answers a different question.

THE HIGH END ~ 99%

Specs that produce very high OTP

- closest snapshot rule
- ignore_early arrivals
- wider lateness window (-2/+7)
- AM peak only

Morning buses run early in light traffic. Combined with **ignore_early**, those early arrivals get classified as on-time.

THE LOW END ~ 20%

Specs that produce low OTP

- last_before snapshot rule
- count_early_as_late
- strict 0/+3 lateness window
- terminals only

Stricter reconstruction, strict symmetric window, treats early as failure, focuses on high-variability stops on a route.

Minimum disclosure checklist for GTFS-RT-based OTP

If you publish OTP from public GTFS-RT data, disclose these alongside the number.

- ✓ **Snapshot-selection rule** — *Often missing*
- ✓ **Lateness window** — *What counts as on time*
- ✓ **Early-arrival treatment** — *Whether early service is a failure*
- ✓ **Stop-selection rule** — *All stops, timepoints, or terminals*
- ✓ **Service-date boundary** — *How midnight-crossing trips are assigned*
- ✓ **Trip-matching rule** — *How real-time updates map to scheduled events*
- ✓ **Time period and service filters** — *Scope of the reported KPI*
- ✓ **Sample size** — *How many observations support the value*
- ✓ **Excluded trips/records** — *Filtering choices (Owl, cancellations)*
- ✓ **GTFS-RT feed type and interval** — *What real-time data is available*
- ✓ **Static GTFS version** — *Schedule benchmark*

Limitations & future work

L I M I T A T I O N S

Two routes, one agency

Routes 23 and 47, SEPTA. Magnitudes may differ across feeds.

Three weeks of data

Apr 4 to Apr 25, 2026. Weekday day-service only.

No AVL ground truth

latest is treated as the closest GTFS-RT-only proxy for actual arrivals — but it remains a proxy.

Six dimensions, not all

Trip matching, missing-data thresholds, weekend/Owl scope could matter.

OTP, not headway

For high-frequency service, headway regularity may better represent rider experience.

F U T U R E W O R K

Cross-agency comparison

Does the snapshot-rule effect generalize beyond SEPTA?

AVL ground-truth comparison

Which GTFS-RT reconstruction rule best approximates actual arrivals?

Headway-based reliability

Apply the same robustness protocol to high-frequency service metrics.

Dashboard/reporting interface

Interactive visualization of the spec curve and disclosure choices.

Reproducible pipeline

Public GitHub repo: [tobibussiek/TransitKPIFramework](#)

T H A N K Y O U

Questions?

OTP cannot be reported as a single unqualified number when computed from public GTFS-RT data.

Disclose how stop events were reconstructed, how on-time was defined, and how sensitive the result is to reasonable alternatives.

Tobias Bussiek

West Chester University of Pennsylvania · M.S. Computer Science

github.com/tobibussiek/TransitKPIFramework

